

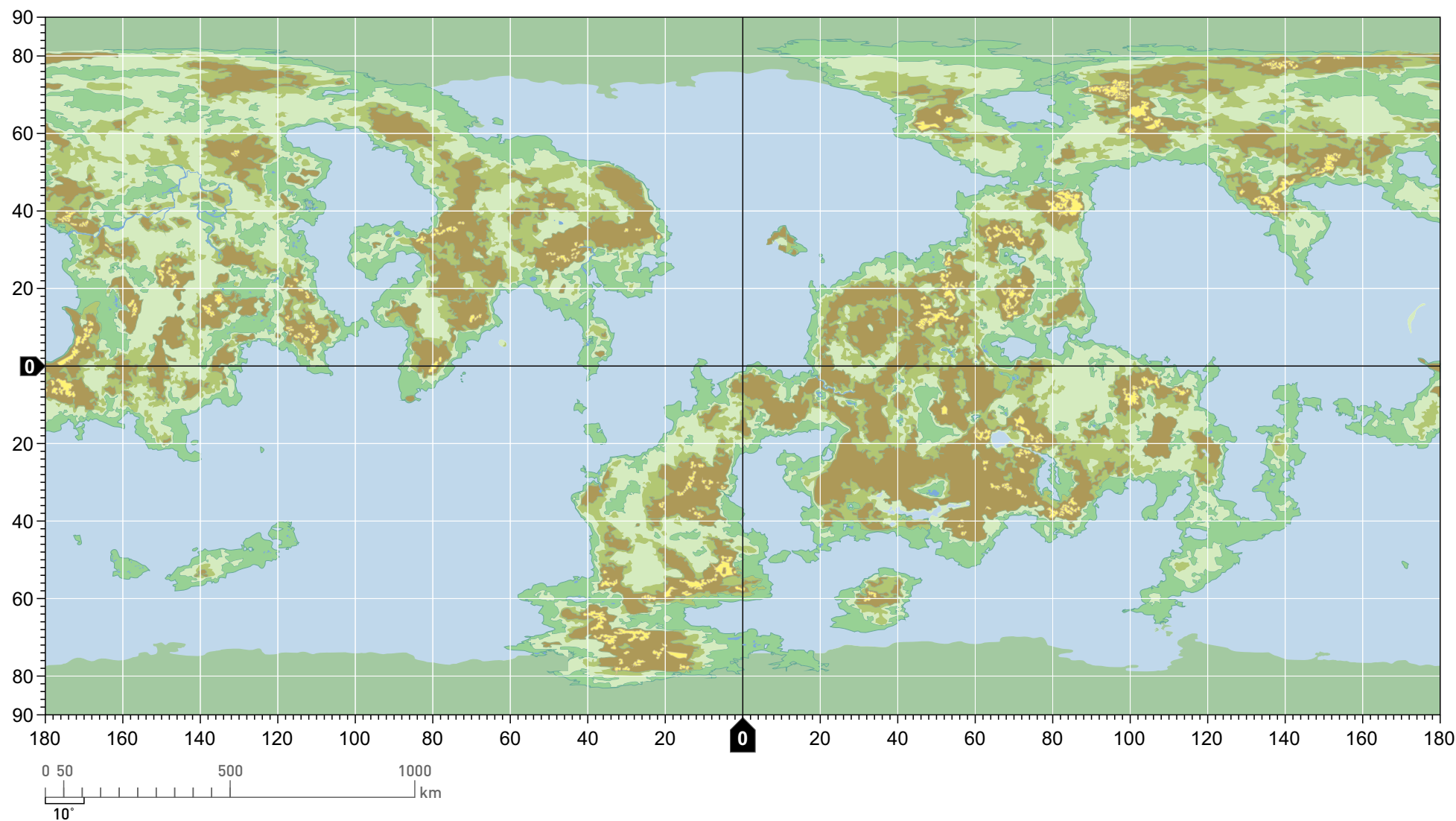


AERONAUTICAL CHARTS

**KERBAL SPACE PROGRAM
KERBIN, 2017**

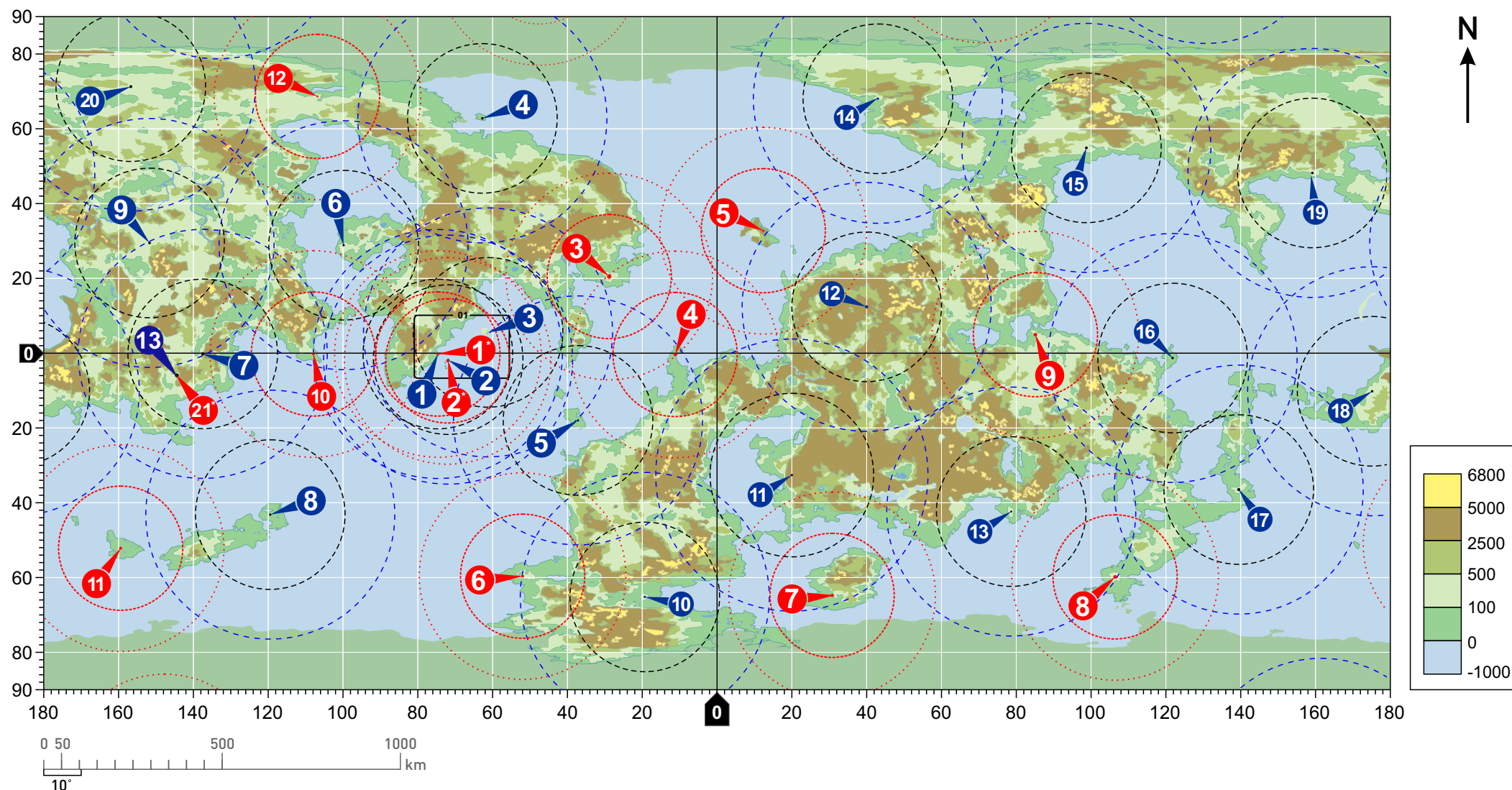
KERBIN

TERRAIN MAP

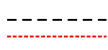


KERBIN

NAVAIDS OVERVIEW



- VOR DME



- range for ALT 5000 m



- detailed view on pages 2-2,2-6,4-1



- NDB DME



- range for ALT 10000 m

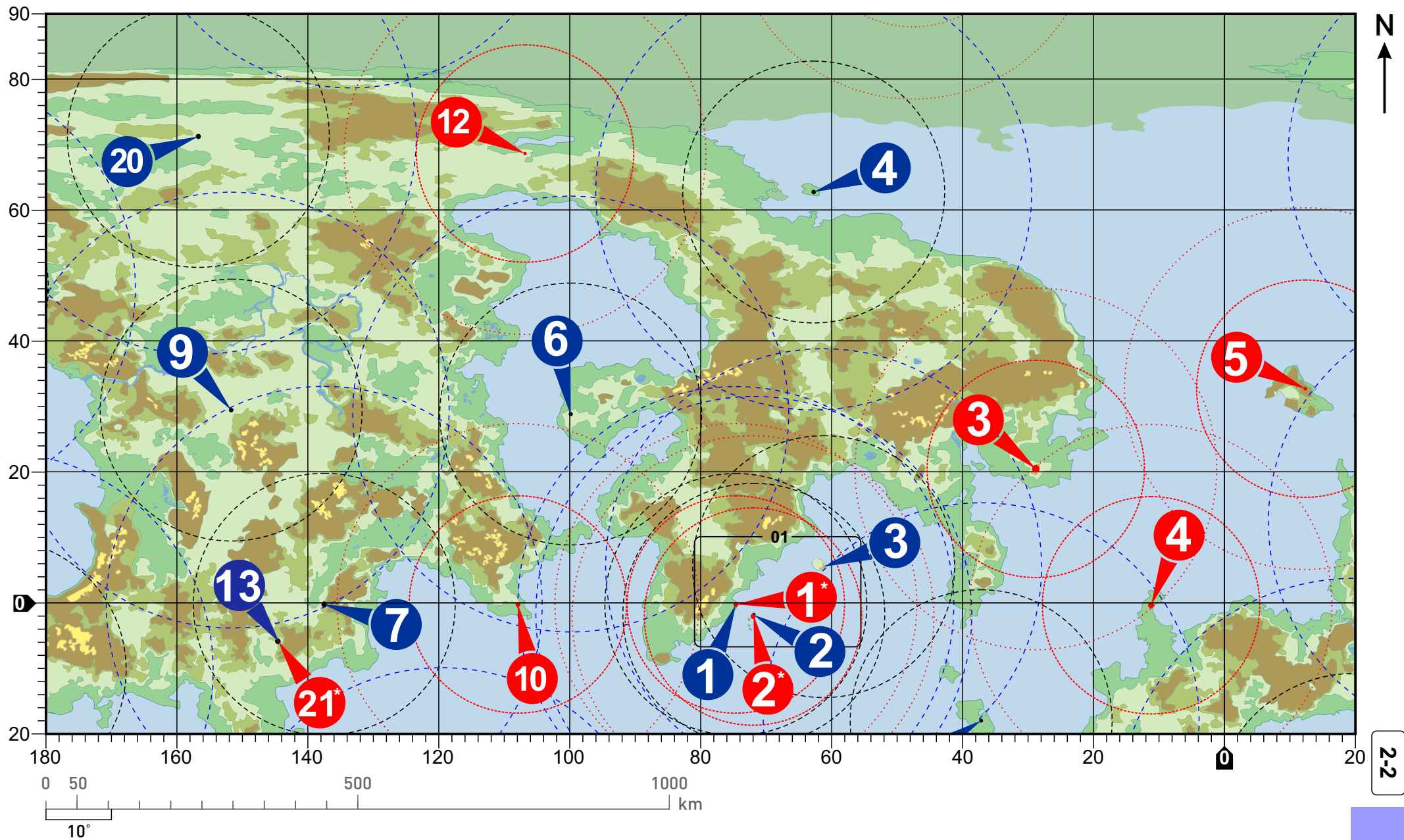


- more than one facilities

	Navigation Aid	VOR	DME	NDB
	approx range for ALT 5000 m	186 KM	217 KM	155 KM
	approx range for ALT 10000 m	263 KM	306 KM	263 KM

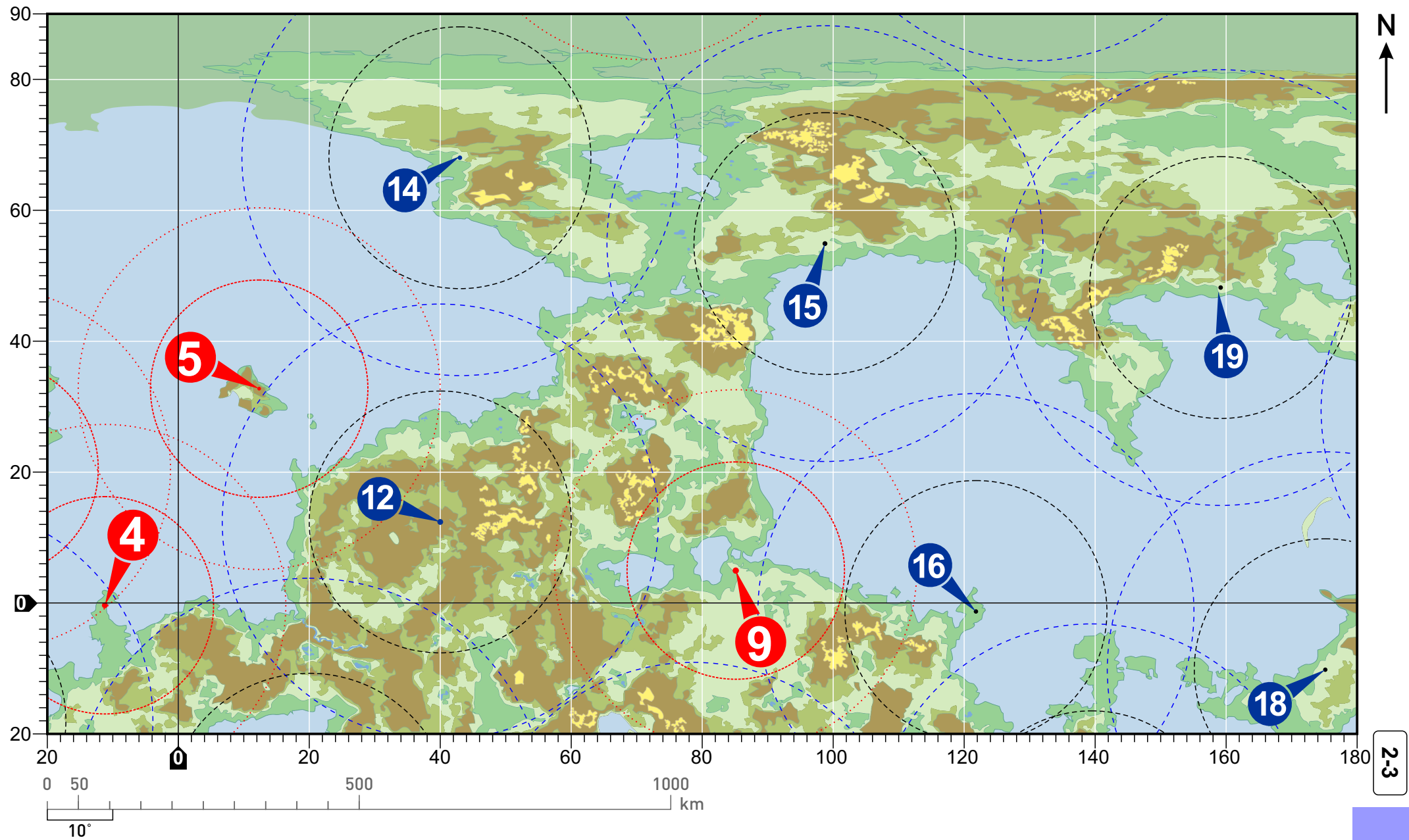
KERBIN

NAVAIDS OVERVIEW (NORTH-WEST)



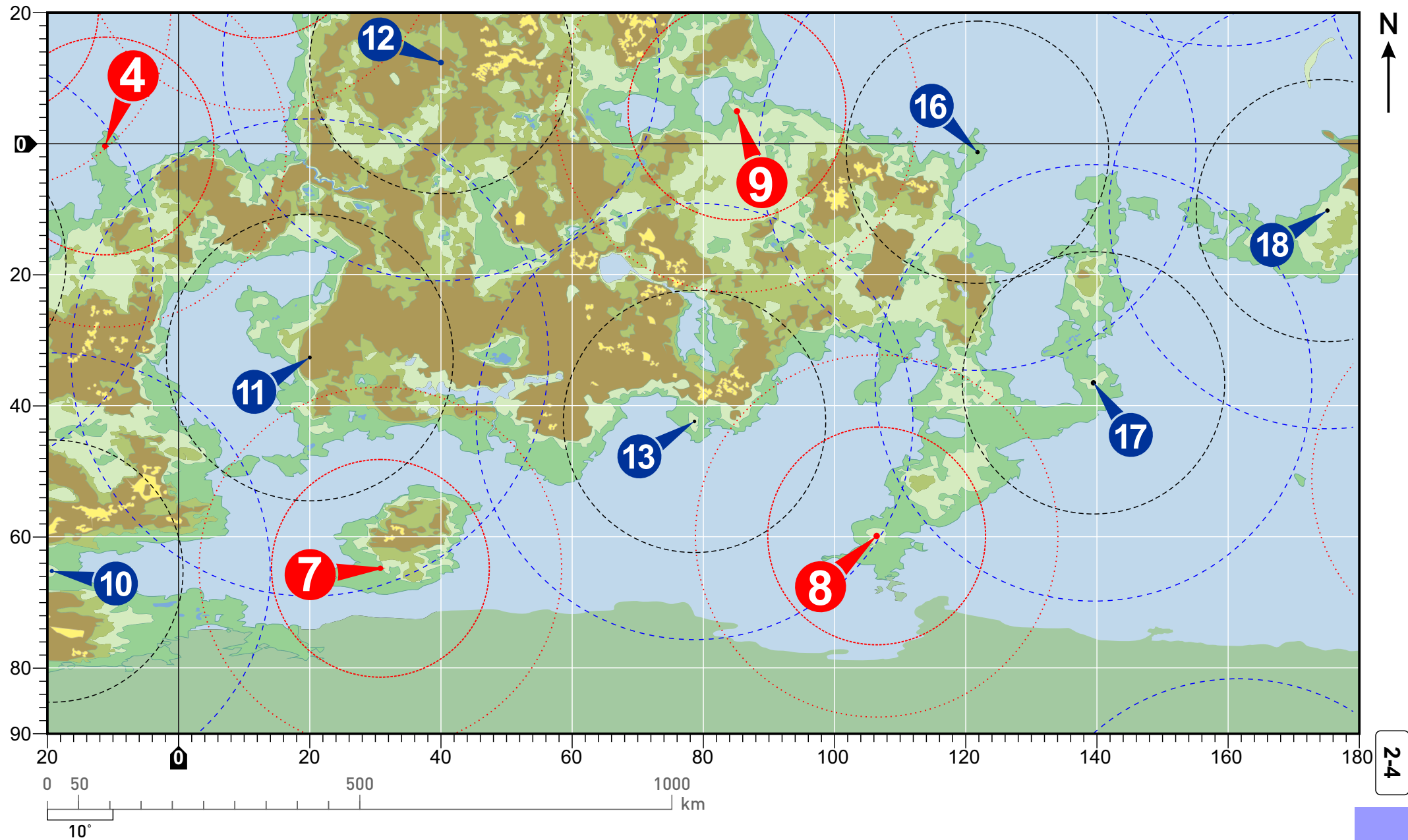
KERBIN

NAVAIDS OVERVIEW (NORTH-EAST)



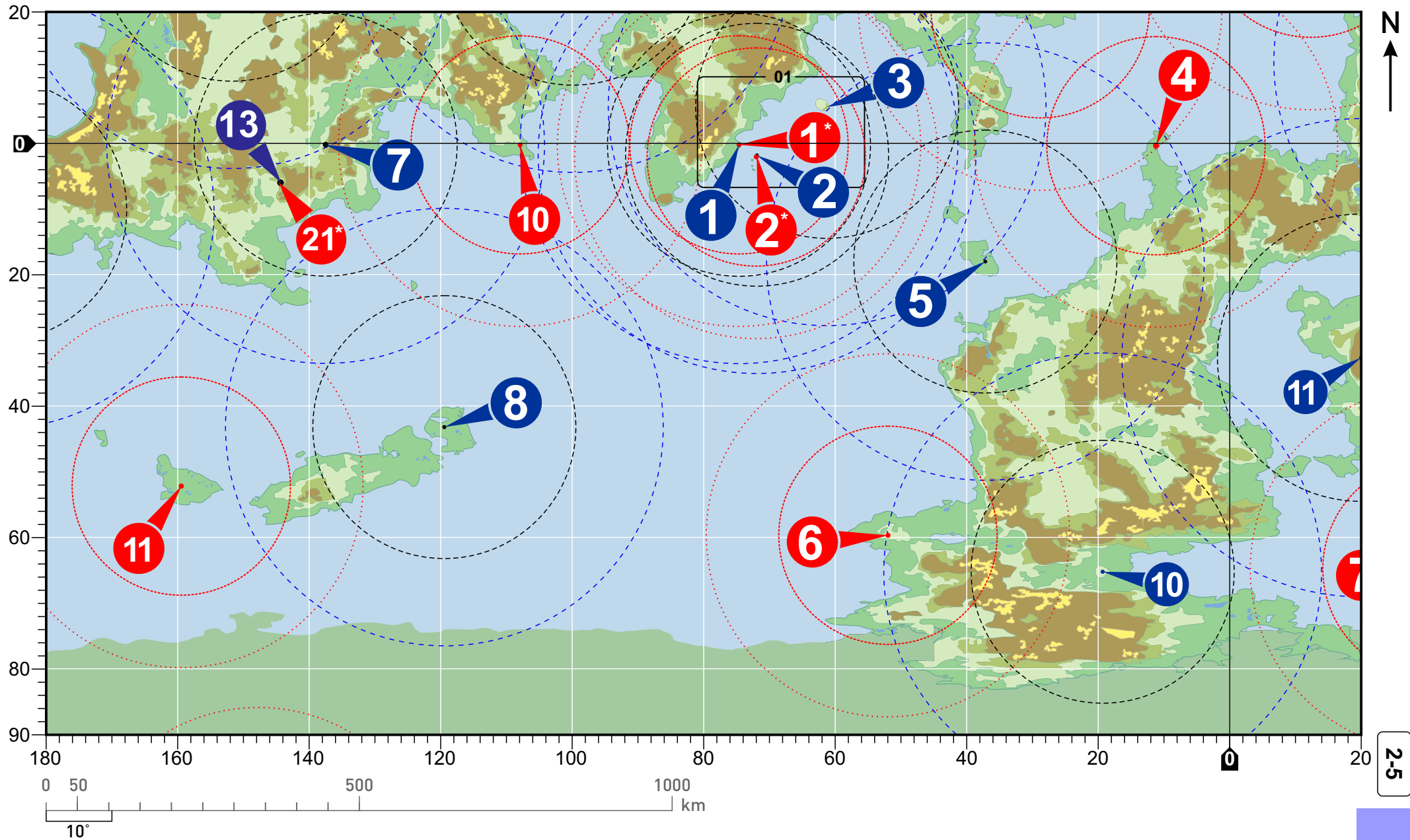
KERBIN

NAVAIDS OVERVIEW (SOUTH-EAST)



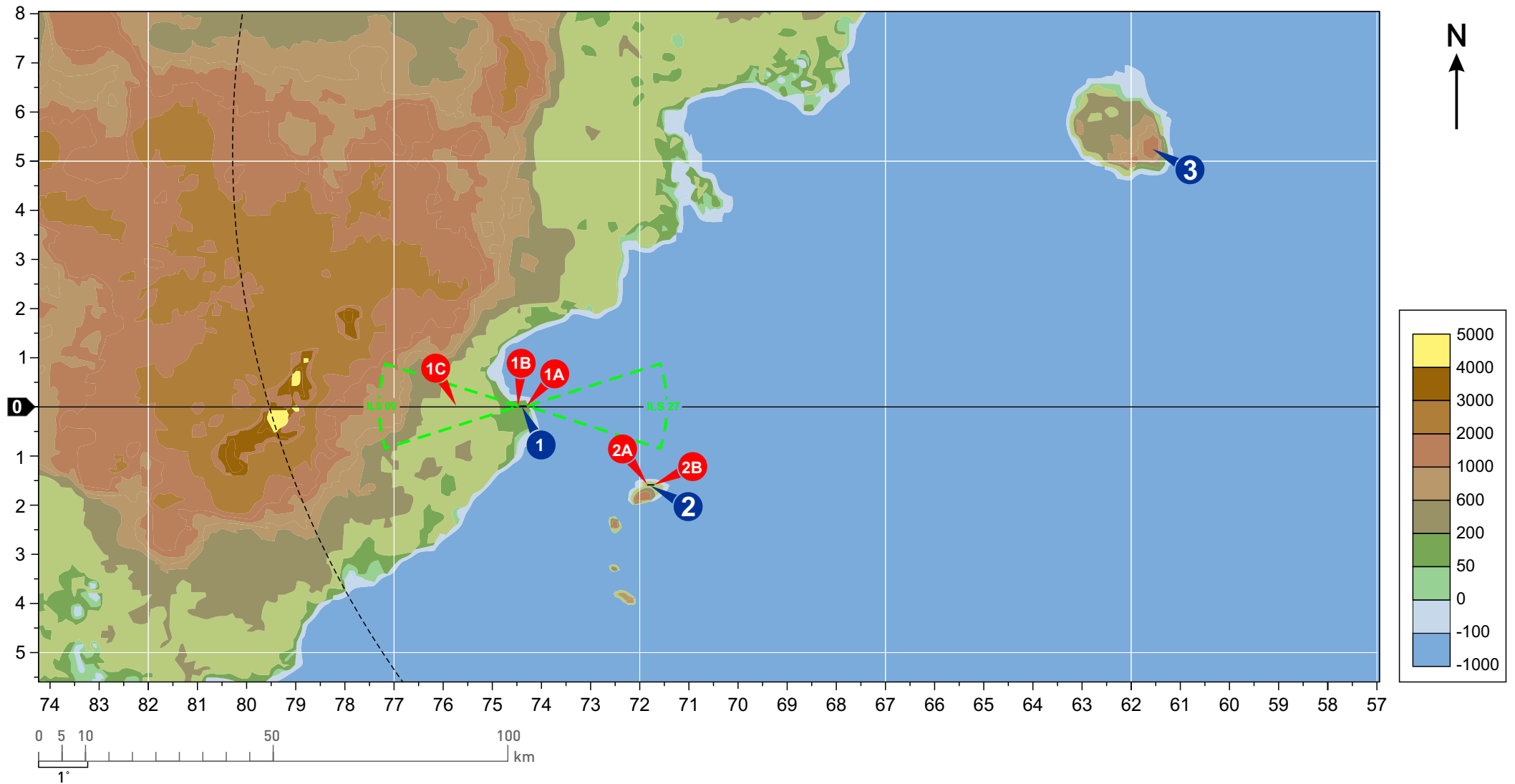
KERBIN

NAVAIDS OVERVIEW (SOUTH-WEST)



KSC SECTOR

NAVAIDS



NAVAIDS

VOR DME

	NAME	ID	FREQUENCY	MORSE CODE	COORDINATES	ALTITUDE
1A	CENTER	KST	113.30	--- ... -	0° 03' 47" S 74° 37' 60" W	160
2	AIRFIELD	IFT	114.20	.. --- ---	1° 31' 24" S 71° 54' 40" W	181.9
3	EASTERN	ESI	116.40		5° 37' 26" S 61° 15' 55" W	1442
4	NAME 04	ID04	112.10	--- ... -	62° 39' 58" N 62° 48' 14" W	414
5	NAME 05	ID05	113.50	--- ... -	18° 19' 31" S 37° 02' 45" W	375
6	NAME 06	ID06	115.10	--- ... -	29° 01' 33" N 99° 45' 21" W	770
7	NAME 07	ID07	114.70	--- ... -	0° 29' 00" S 136° 33' 36" W	1462
8	NAME 08	ID08	116.70	--- ... -	43° 00' 02" S 120° 19' 20" W	273
9	NAME 09	ID09	113.80	--- ... -	30° 04' 50" N 151° 27' 27" W	760
10	NAME 10	ID10	115.20	--- ... -	64° 49' 09" S 19° 14' 53" W	825
11	NAME 11	ID11	117.10	--- ... -	31° 33' 10" S 21° 47' 49" E	2136
12	NAME 12	ID12	114.30	--- ... -	12° 09' 03" N 39° 52' 49" E	1323
13	NAME 13	ID13	117.60	--- ... -	44° 50' 09" S 78° 35' 46" E	889
14	NAME 14	ID14	113.10	--- ... -	70° 22' 42" N 41° 25' 07" E	1137
15	NAME 15	ID15	116.60	--- ... -	54° 17' 40" N 99° 42' 43" E	632
16	NAME 16	ID16	115.30	--- ... -	1° 19' 06" S 121° 46' 21" E	359
17	NAME 17	ID17	117.20	--- ... -	36° 45' 37" S 139° 21' 54" E	596

[illegible]

NAVAIDS

NDB DME

	NAME	ID	FREQUENCY	MORSE CODE	COORDINATES	ALTITUDE		ID	FREQUENCY	MORSE CODE	COORDINATES	ALTITUDE	
1A	KKSC ILS 27	IWE	110.10	.. --- .	0° 03' 01" S 74° 31' 03" W	70	13C	DHLAP	SCS	205.00	... -.-. ...	9° 27' 33" S 144° 5' 29" W	501
1B	KKSC ILS 09	IKW	108.50	.. --- ---	0° 02' 55" S 74° 41' 57" W	70	13D	DTAKE	SCN	206.00	... -.-. -.	3° 35' 16" S 143° 59' 18" W	842
1C	BAKOR	SCE	197.00	... --- .	0° 02' 55" S 77° 33' 54" W	745							
2A	BASE WEST	IRW	191.00	.. --- ---	1° 30' 58" S 71° 58' 53" W	83.4							
2B	BASE EAST	IRE	194.00	.. --- .	1° 30' 58" S 71° 50' 18" W	97.8							
3	NAME 03	Id03	192.00	--- ... -	20° 41' 54" N 28° 47' 03" W	1039							
4	NAME 04	Id04	201.00	--- ... -	0° 46' 08" N 10° 24' 54" W	228							
5	NAME 05	Id05	208.00	--- ... -	32° 24' 35" N 12° 34' 06" E	2208							
6	NAME 06	Id06	211.00	--- ... -	59° 22' 12" S 52° 11' 06" W	530							
7	NAME 07	Id07	209.00	--- ... -	64° 22' 48" S 31° 35' 48" E	518							
8	NAME 08	Id08	195.00	--- ... -	60° 09' 40" S 106° 23' 30" E	605							
9	NAME 09	Id09	198.00	--- ... -	4° 02' 35" N 85° 44' 14" E	476							
10	NAME 10	Id10	202.00	--- ... -	0° 15' 49" N 108° 37' 58" W	514							
11	NAME 11	Id11	203.00	--- ... -	51° 57' 55" S 159° 27' 20" W	340							
12	NAME 12	Id12	210.00	--- ... -	68° 25' 22" N 108° 28' 44" W	747							
13A	KDAF ILS 36	IKS	110.30	.. --- ...	6° 35' 41" S 144° 2' 26" W	821							
13B	KDAF ILS 18	ISN	108.70	 -.	6° 27' 8" S 144° 2' 18" W	821							

NAVAIDS
ILS DME AND MARKERS

NAME ID FREQUENCY MORSE CODE COORDINATES ALTITUDE TDZ
DISTANCE

KERBAL SPACE CENTER (KKSC)

DESSERT AIR FIELD (KDAF)

RWY 09

RWY 18

1	KKSC ILS 09	IKW	108.50	.. --- ---	0° 02' 55" S 74° 41' 57" W	70	
	Outer Marker				0° 02' 55" S 75° 29' 31" W	240	8300
	Middle Marker				0° 02' 55" S 74° 49' 26" W	65	1300
	Inner Marker				0° 02' 55" S 74° 44' 49" W	65	500

KDAF ILS 18	ISN	108.70	 -.	6° 27' 8" S 144° 2' 18" W	820	
Outer Marker				5° 39' 35" S 144° 1' 28" W	593	8300
Middle Marker				6° 2' 30" S 144° 1' 52" W	610	4300
Inner Marker				6° 24' 16" S 144° 2' 15" W	820	500

RWY 27

RWY 36

2	KKSC ILS 27	IWE	110.10	.. --- .	0° 03' 01" S 74° 31' 03" W	70	
	Middle Marker				0° 03' 01" S 74° 23' 38" W	0	1100
	Inner Marker				0° 03' 01" S 74° 28' 10" W	57	500

KDAF ILS 36	IKS	110.30	.. --- ...	6° 35' 41" S 144° 2' 26" W	820	
Outer Marker				7° 23' 14" S 144° 3' 16" W	692	8300
Middle Marker				7° 0' 19" S 144° 2' 52" W	991	4300
Inner Marker				6° 38' 33" S 144° 2' 29" W	820	500

ISLAND AIR FIELD (KAFD)

RWY 09

3	Inner Marker				1° 30' 58" S 71° 50' 18" W	98	146
---	--------------	--	--	--	-------------------------------	----	-----

RWY 27

4	Inner Marker				1° 30' 58" S 71° 51' 19" W	134	155
---	--------------	--	--	--	-------------------------------	-----	-----

JEBBESEN
25-Nov-2017

ELEVATIONS AND
DIMENSIONS IN METERS

RWY	DIRECTION	THR
09	89°51'	0°03'01"S 74°29'19" W
27	270°09'	0° 02' 55" S 74°43'42" W

3-1

AERODROME
CHART

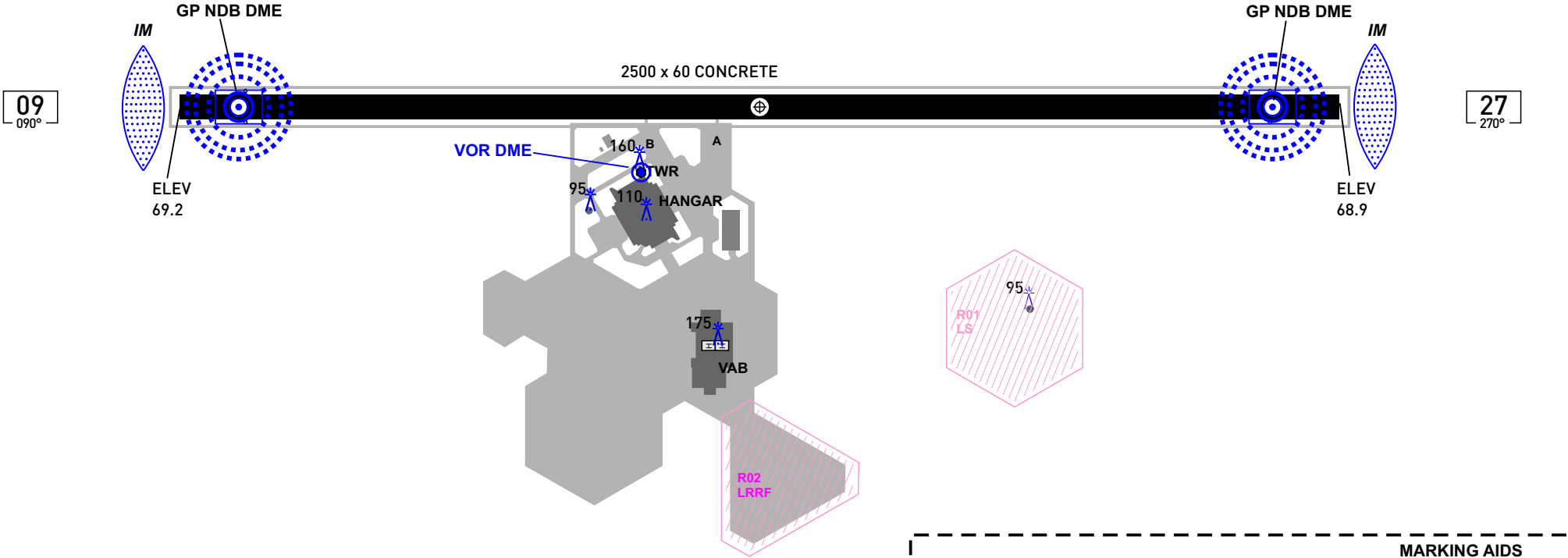
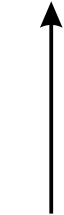
0°03'00"S
74°36'27" W

ELEV
69.1

TOWER 122.00
GROUND 121.70

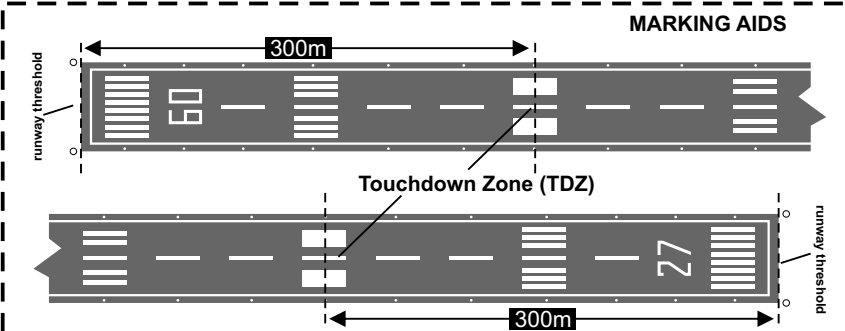
KKSC

3-1



09
090°

27
270°



JEBBESEN

25-Nov-2017

ELEVATIONS AND
DIMENSIONS IN METERS

RWY

DIRECTION

THR

09

90°00'

1° 30' 58" S
71° 58' 02" W

27

270°00'

1° 30' 58" S
71° 51' 19" W

3-2

AERODROME
CHART

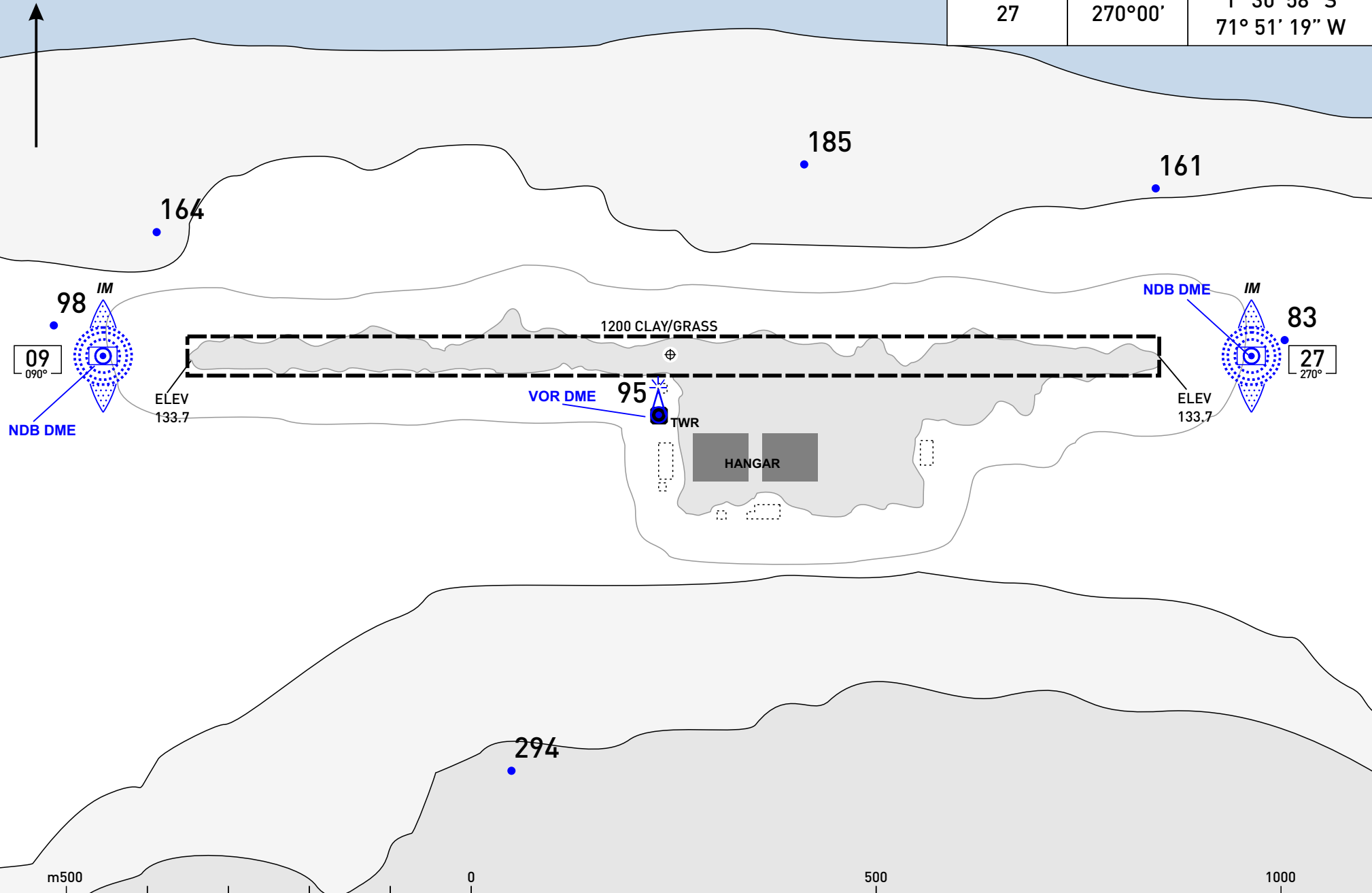
1° 31' 03" S
71° 54' 40" W

ELEV
133.7

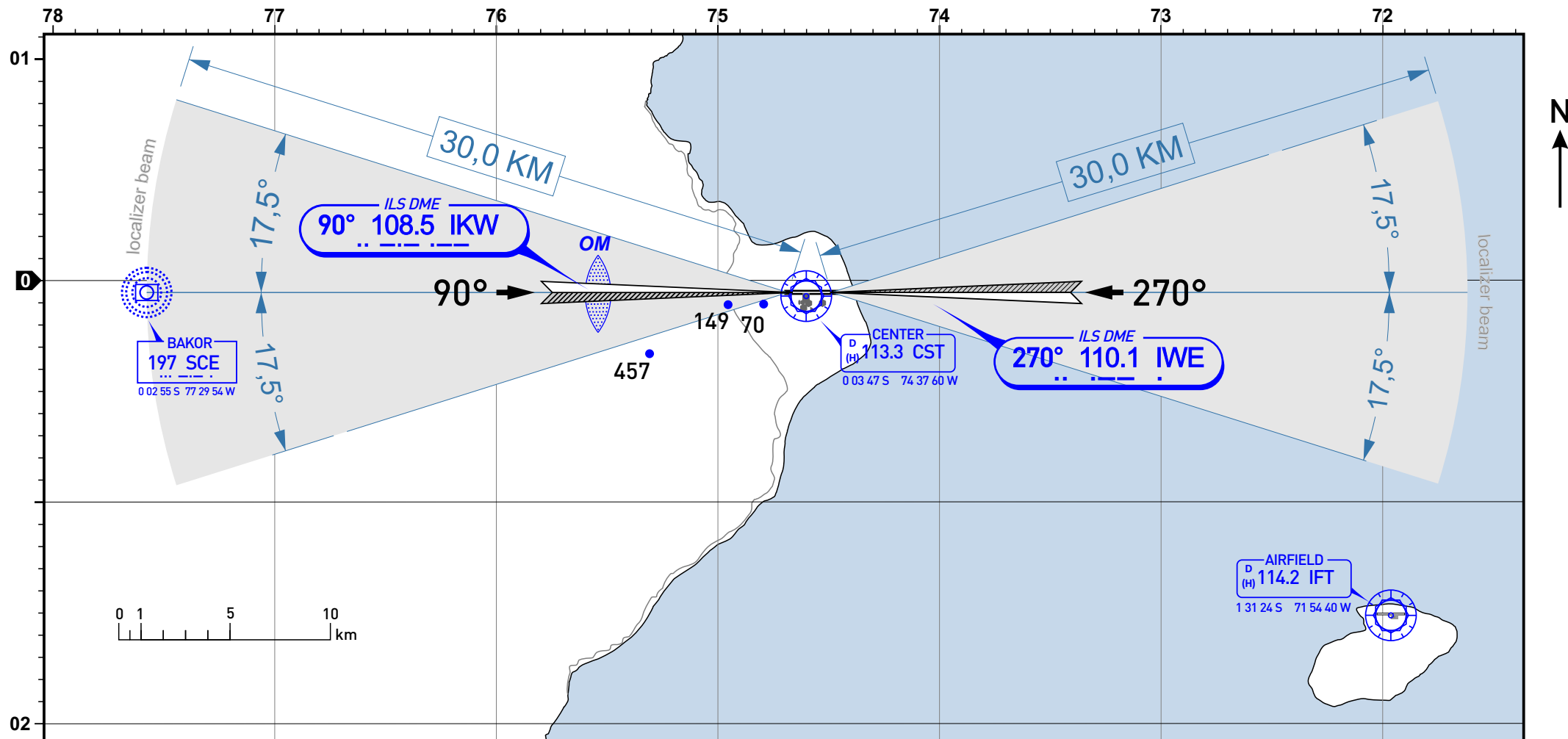
TOWER 122.00
GROUND 121.70

KERBIN
KAFD

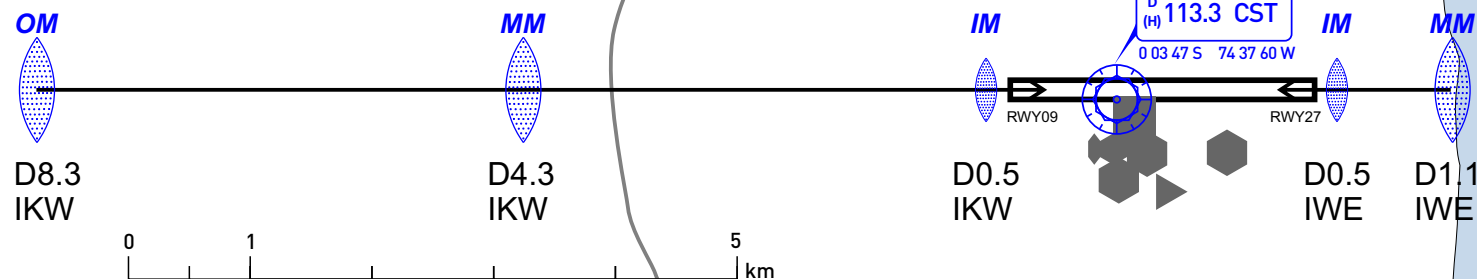
3-2



KKSC ILS 09, ILS 27 OVERVIEW



MARKERS



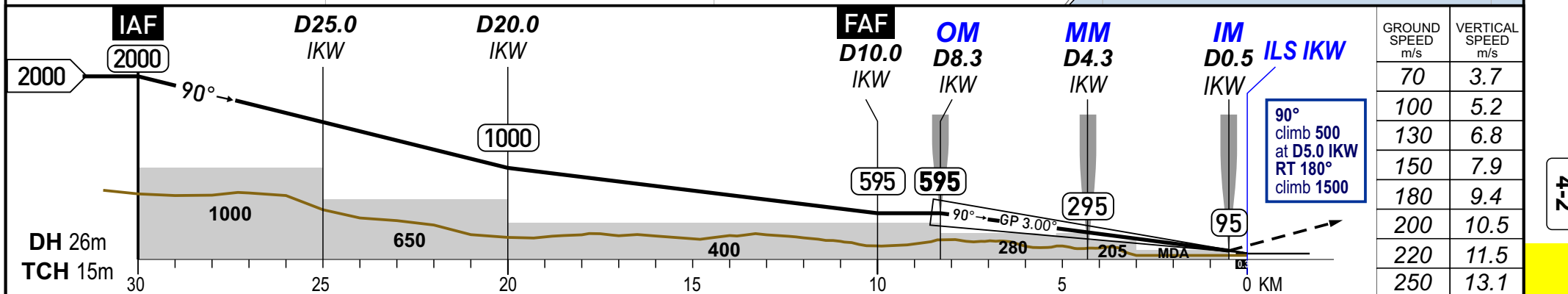
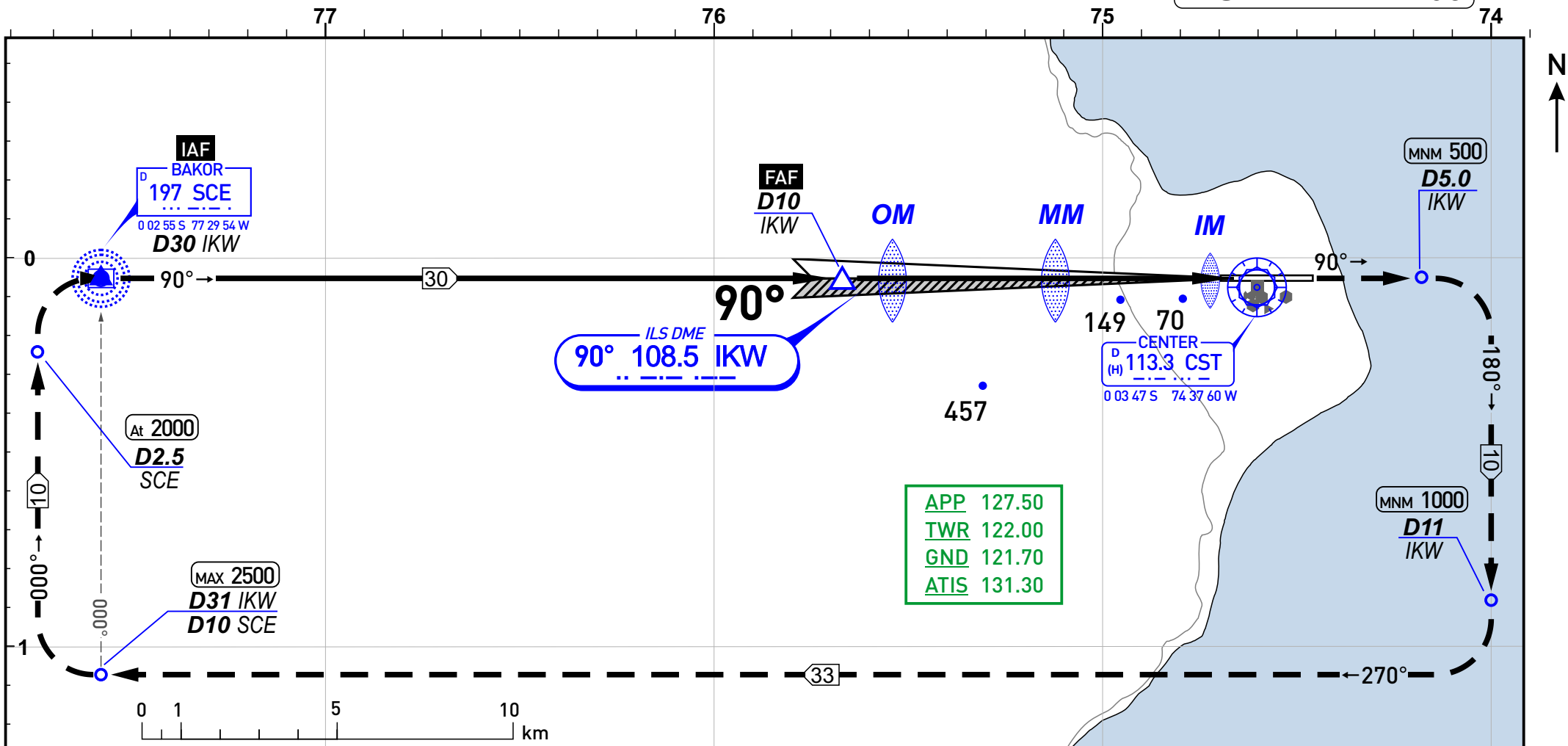
JEBBESEN

25-Nov-2017

KKSC

4-2

ILS DME RWY 09



4-2

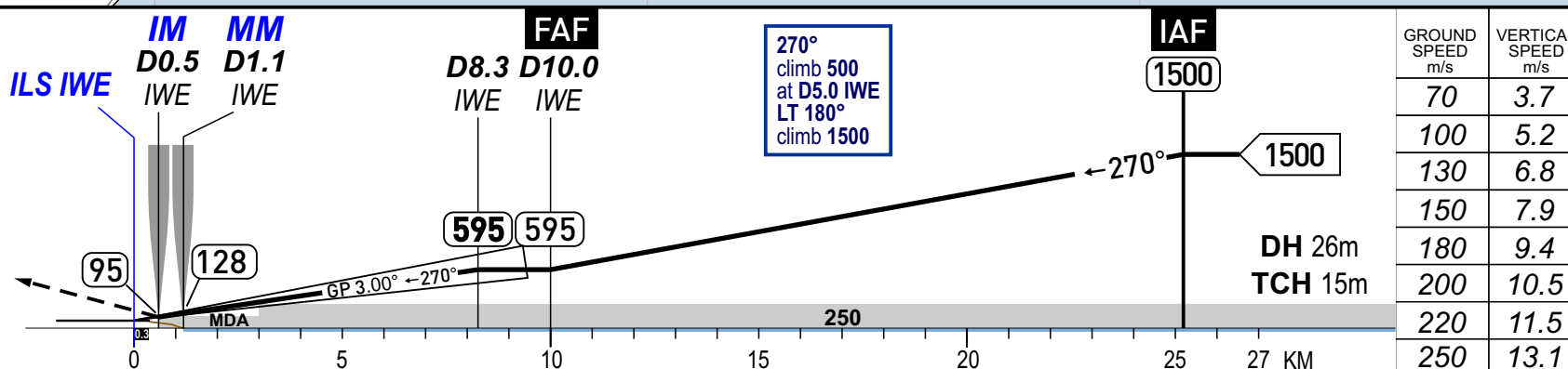
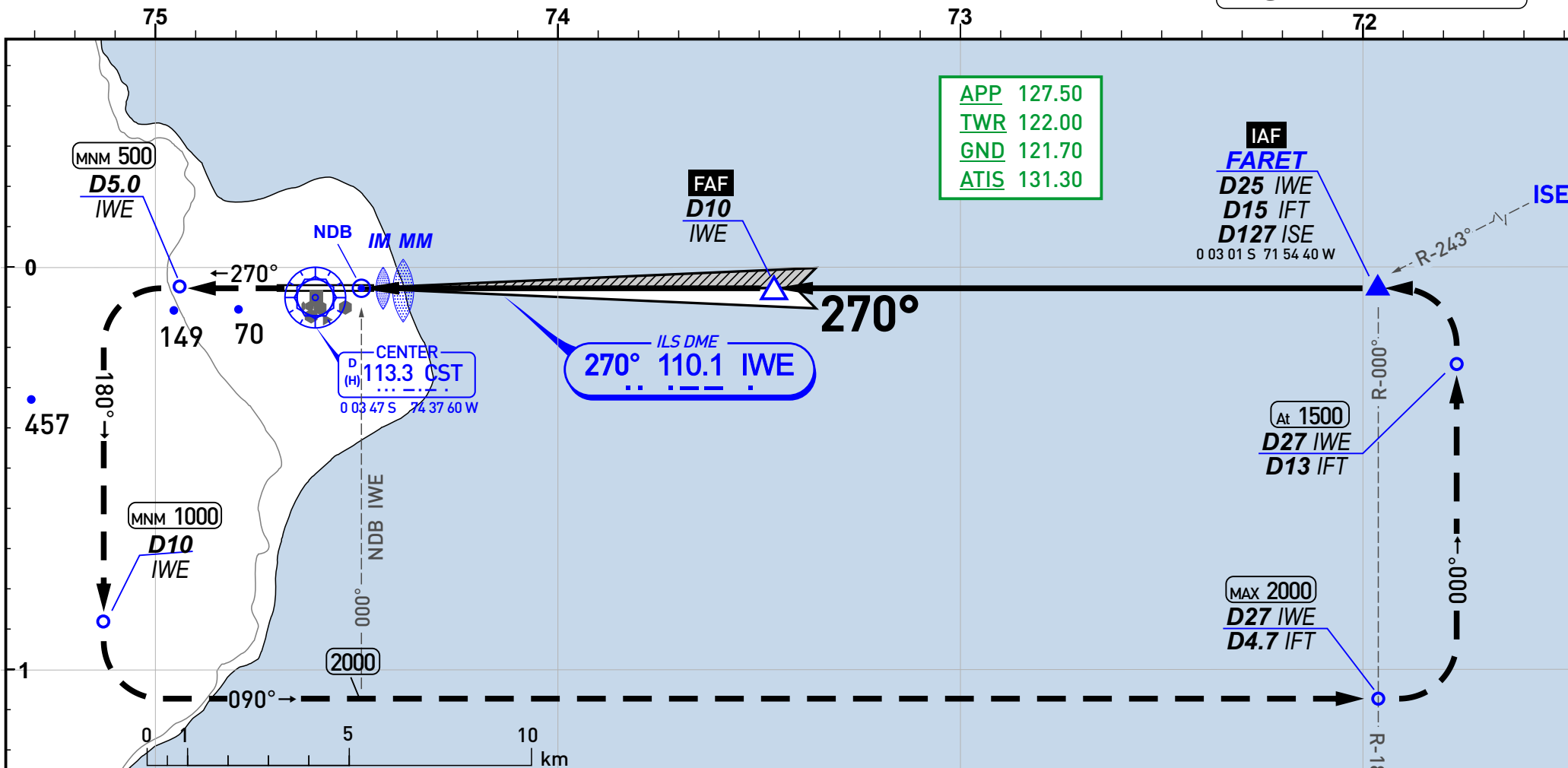
JEBBESEN

25-Nov-2017

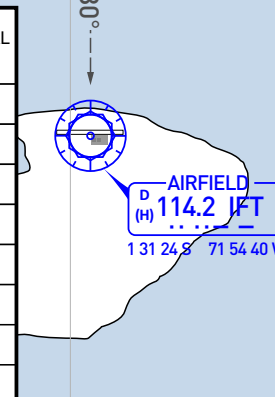
KKSC

4-3

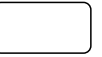
ILS DME RWY 27



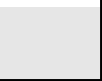
GROUND SPEED m/s	VERTICAL SPEED m/s
70	3.7
100	5.2
130	6.8
150	7.9
180	9.4
200	10.5
220	11.5
250	13.1



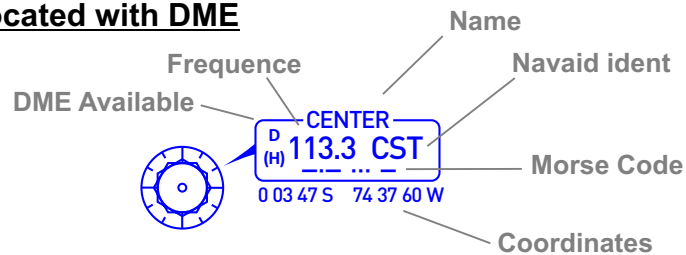
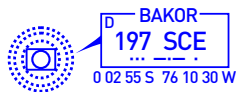
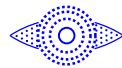
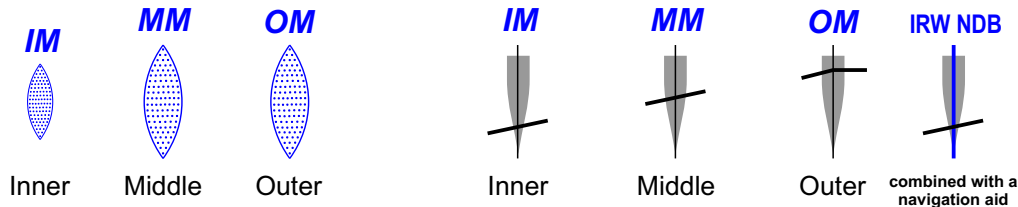
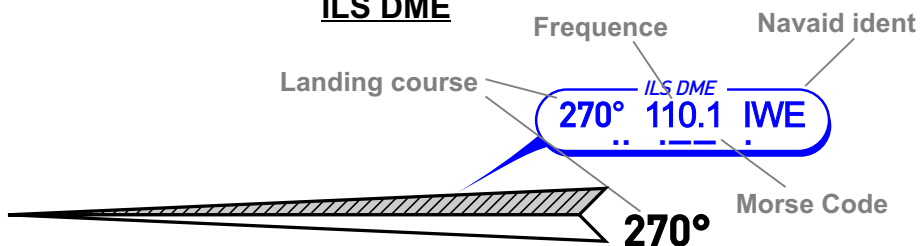
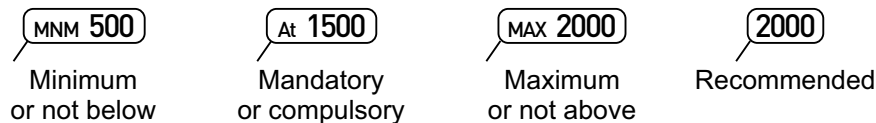
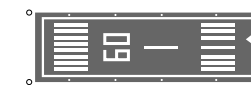
4-3



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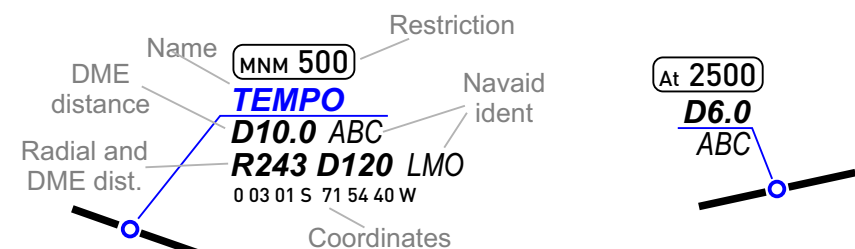
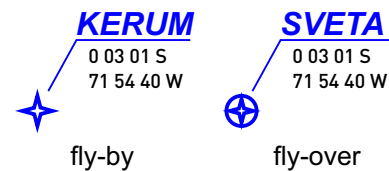
AEROCHART LEGEND

VOR co-located with DMENDB co-located with DMENDB co-located with markerThe symbol for a nav aidPlan viewMarkersProfile viewILS DMEAltitudesStructureTerrain elevationHighest terrain pointRestricted and danger areasWater areaAerodrome symbolsClosed
AerodromeHelicopter landing siteRunways

Paved



Unpaved

Waypoint / Fix (with radio aid)Waypoint / Fix (RNAV/GPS)Mileage breakApproach fix

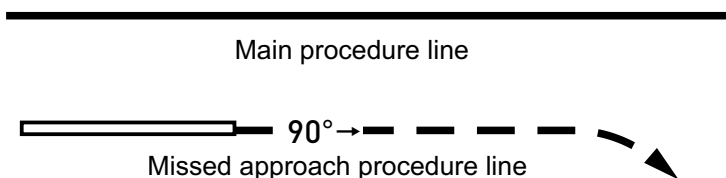
Initial



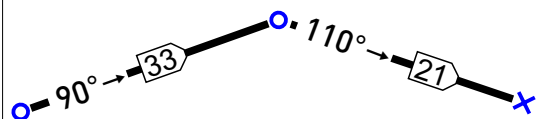
Final

AEROCHART LEGEND

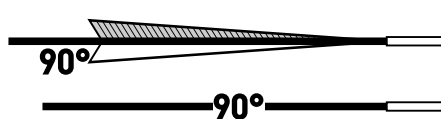
Procedure lines



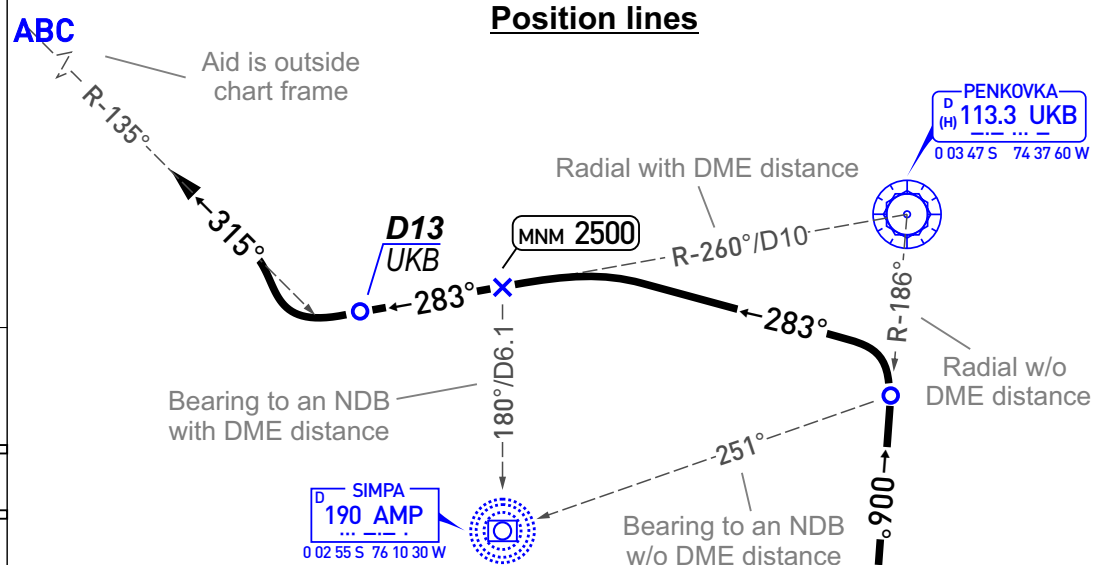
Track and distance box



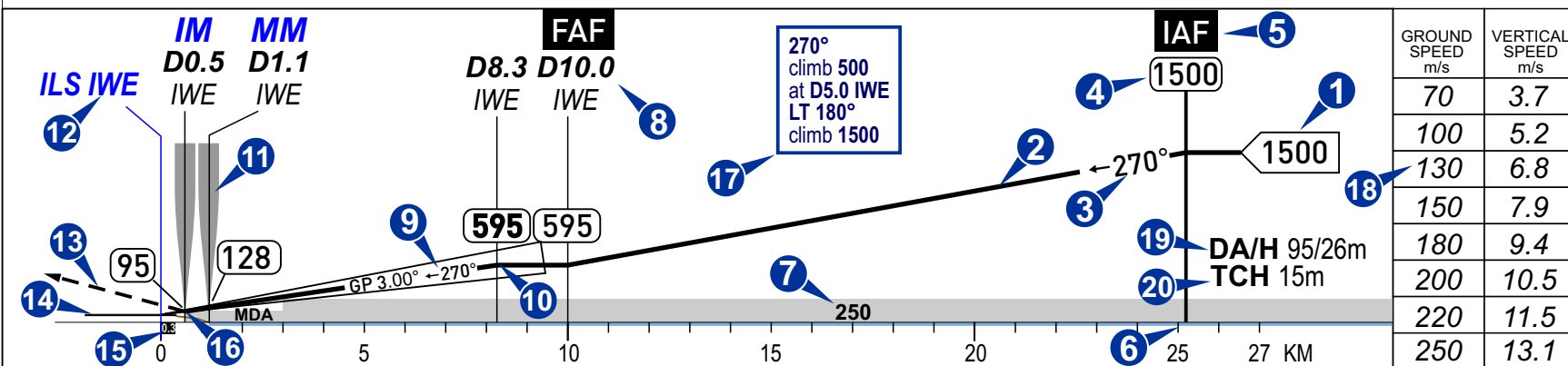
Final approach track.



Position lines



Profile view



- Altitude before starting final approach
- Procedure line
- Average approach angle
- Minimum altitude over or abeam a facility or fix
- Initial approach fix - IAF
- Distance bar referring to Touchdown Zone (TDZ)

- Minimum Safe Altitudes
- Intersection defined by a DME distance
- ILS glide path symbol and angle (if available)
- Glide slope intercept
- Marker combined with an intersection (defined by a DME distance)
- Navigation aid
- Missed approach procedure symbol
- Runway symbol
- Distance from runway threshold to TDZ
- Missed approach point (MAPt).
- Missed approach procedure text
- The table creates a constant rate of descent (3.00°)
- Decision altitude/Height (DA/H)
- Threshold crossing height (TCH)

DEFINITIONS

Aerodrome elevation. The elevation of the highest point of the landing area.

Altitude. The vertical distance of a level, a point or an object considered as a point, measured above sea level (ASL).

Area navigation (RNAV). A method of navigation which permits aircraft operation on any desired flight path within the coverage of the station-referenced navigation aids or within the limits of the capability of self-contained aids, or a combination of these.

Decision altitude (DA) or decision height (DH). A specified altitude or height at which a missed approach must be initiated if the required visual reference to continue the approach has not been established. *Decision altitude (DA)* is referenced to ASL and *decision height (DH)* is referenced to the AGL (Above ground level)

DME distance. The line of sight distance (slant range) from the source of a DME signal to the receiving antenna.

Heading. The direction in which the longitudinal axis of an aircraft is pointed, usually expressed in degrees from North.

Initial approach fix (IAF). A fix that marks the beginning of the initial segment and the end of the arrival segment, if applicable. In RNAV applications this fix is normally defined by a fly-by waypoint.

Minimum descent altitude (MDA). A specified altitude in approach below which descent must not be made without the required visual reference. Minimum descent altitude (MDA) is referenced to ASL

Missed approach point (MAPt). That point in an approach procedure at or before which the prescribed missed approach procedure must be initiated in order to ensure that the minimum obstacle clearance is not infringed.

Missed approach procedure. The procedure to be followed if the approach cannot be continued.

Threshold (THR). The beginning of that portion of the runway usable for landing.

Track. The projection on the earth's surface of the path of an aircraft, the direction of which path at any point is usually expressed in degrees from North.

Waypoint. A specified geographical location used to define an area navigation route or the flight path of an aircraft employing area navigation. Waypoints are identified as either:

Fly-by waypoint. A waypoint which requires turn anticipation to allow tangential interception of the next segment of a route or procedure, or

Flyover waypoint. A waypoint at which a turn is initiated in order to join the next segment of a route or procedure.

ABBREVIATIONS
AND ACRONYMS

AGL	Above ground level
ASL	Above sea level
ATC	Air traffic control
CDI	Course deviation indicator
DA/H	Decision altitude/height
DME	Distance measuring equipment
EFIS	Electronic flight instrument system
FAF	Final approach fix
FMC	Flight management computer
GNSS/	Global navigation satellite system
GPS	
GP	Glide path
GPWS	Ground proximity warning system
HSI	Horizontal situation indicator
IAS	Indicated airspeed
ILS	Instrument landing system
kt	Knot(s)
km	Kilometre(s)
RNAV	Lateral navigation
m	Metre(s)
MAPt	Missed approach point
MDA/H	Minimum descent altitude/height
NDB	Non-directional beacon
NM	Nautical mile(s)
OM	Outer marker
RNAV	Area navigation
THR	Threshold
VNAV	Vertical navigation
VOR	Very high frequency omnidirectional radio range
VPA	Vertical path angle

SPEED
CONVERSION

SPEED m/s	km/min	km/h
5	0,3	18
6	0,36	21,6
7	0,42	25,2
8	0,48	28,8
9	0,54	32,4
10	0,6	36
11	0,66	39,6
12	0,72	43,2
13	0,78	46,8
14	0,84	50,4
15	0,9	54
16	0,96	57,6
17	1,02	61,2
18	1,08	64,8
19	1,14	68,4
20	1,2	72
21	1,26	75,6
22	1,32	79,2
23	1,38	82,8
24	1,44	86,4
25	1,5	90
26	1,56	93,6
27	1,62	97,2
28	1,68	100,8
29	1,74	104,4
30	1,8	108
31	1,86	111,6
32	1,92	115,2
33	1,98	118,8
34	2,04	122,4
35	2,1	126
36	2,16	129,6
37	2,22	133,2
38	2,28	136,8
39	2,34	140,4
40	2,4	144
41	2,46	147,6
42	2,52	151,2
43	2,58	154,8
44	2,64	158,4
45	2,7	162
46	2,76	165,6
47	2,82	169,2
48	2,88	172,8
49	2,94	176,4
50	3	180
55	3,3	198
60	3,6	216
65	3,9	234

NORMAL OPERATING CHECKLISTS

BEFORE ENGINE START

PARKING BRAKE.....	SET
THROTTLE.....	IDLE
LANDING GEAR.....	DOWN-GREEN
FLAPS.....	RETRACTED
LIGHTS.....	ON
FLIGHT PLAN.....	FILED
STAGING.....	CHECK
FUEL QUANTITY.....	CHECK
BATTERY CHARGE.....	CHECK

DESCENT

BRIEFING.....	CONFIRMED
NAV RADIO.....	SET
LIGHTS.....	ON
APU.....	ON
AUTO SPEED BRAKE.....	ARM

APPROACH / LANDING

ENGINE GIMBAL.....	AS REQUIRED
ENGINE MODE.....	AS REQUIRED
THRUST LIMIT.....	AS REQUIRED
NAV RADIO.....	SET
RADAR ALT.....	MDA/DH SET
LANDING GEAR.....	DOWN-GREEN
FLAPS.....	AS REQUIRED
TRIM.....	AS REQUIRED
CABIN.....	READY

AFTER LANDING

AUTO SPEED BRAKE.....	OFF
SPOILER.....	RETRACTED
PARKING BRAKE.....	OFF
SAS.....	OFF

BEFORE TAKEOFF

BRIEFING.....	CONFIRMED
APU.....	ON
AVIONICS.....	CHECK
SAS.....	ON, STAB
ENGINE GIMBAL.....	AS REQUIRED
ENGINE MODE.....	AS REQUIRED
THRUST LIMIT.....	AS REQUIRED
NAV RADIO.....	SET
LIGHTS.....	ON
PARKING BRAKE.....	SET
BRAKES FORCE.....	SET
AUTO SPEED BRAKE.....	OFF
SPOILER.....	RETRACTED
FLAPS.....	AS REQUIRED
TRIM.....	AS REQUIRED
FLT CONTROLS.....	TEST
CABIN.....	READY
ATC.....	ADVISE

AFTER TAKEOFF

FLAPS.....	RETRACTED
LANDING GEAR.....	UP
LIGHTS.....	OFF
APU.....	OFF

PARKING

PARKING BRAKE.....	SET
LIGHTS.....	OFF
TRIM.....	RESET-CHECK
THROTTLE.....	IDLE
ENGINE.....	SHUTDOWN
STAGING.....	CHECK
VESSEL RECOVERY CALL.....	AS REQUIRED

